

31 Scatter Graphs

- | | |
|--|--|
| 1. 0.36cm | 2. 1.44cm ² |
| 3. 27, 17, 7, -3, -13, -23 | 4. $0.1^3 = 0.001$ |
| 5. $\sqrt[3]{81} \approx 9$ | 6. A right angle = 90° |
| 7. A sample of every nth person is called a <i>systematic</i> sample | 8. Putting all the names in a hat is called a <i>random</i> sample |
| 9. 10 | 10. 0.4 |

Activity: Body Ratios

This activity gives an introduction to bivariate data analysis and an opportunity to reinforce the statistical investigation process.

Each individual student should draw a graph on graph paper and put trend line in and compare their results to one that is drawn by teacher (by hand, or by technology).

32 Probability Expressions

- | | |
|---|---|
| 1. 0.2 | 2. $\frac{1}{4}$ |
| 3. $\frac{1}{4}$ | 4. 5% |
| 5. $\frac{1}{4} \times \frac{1}{4} = \frac{1}{8}$ | 6. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ |
| 7. $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$ | 8. $(0.5)^2 = 0.25$ |
| 9. $(0.05)^2 = 0.0025$ | 10. $(0.01)^2 = 0.0001$ |

33 Simulations

- | | |
|------------------------------------|------------------------------------|
| 1. 8 | 2. IIII IIII III |
| 3. $\frac{1}{2}$ | 4. $\frac{1}{6}$ |
| 5. 0 | 6. 1 |
| 7. $1 - \frac{2}{3} = \frac{1}{3}$ | 8. $1 - \frac{2}{5} = \frac{3}{5}$ |
| 9. less likely | 10. more likely |

Class Activity I: Infection simulation

One student will start as infected, give this person the 0 worksheet, the rest of the students get the 1 worksheet. Let the students mingle for a minute or two before stopping them. At each interval ask how many 0's are in the class and record this on the board. Keep this information for the final discussion.

Trial	Your Number
1	0
2	
3	
4	
5	

Trial	Your Number
1	1
2	
3	
4	
5	

Trial	Your Number
1	2
2	
3	
4	
5	

Trial	Your Number
1	3
2	
3	
4	
5	

Name: _____

Year 12 Essentials



Statistical thinking process weekly task

You will be presented with bivariate data to interpret and analyse. You will need to draw scatterplots (it is up to you if you want to do this by hand or with technology) and use these to write about any patterns/features with the data. You should then be able to conclude if there is an association with the data.

You will need to apply the statistical thinking process:

Part A: clarify the problem and pose one or more questions that can be answered with data

Part B: design and implement a plan to collect or obtain appropriate data

Part C: select and apply appropriate graphical or numerical techniques to analyse the data

Part D: interpret the results of this analysis and relate the interpretation to the original question

Part E: communicate findings in a systematic and concise manner.

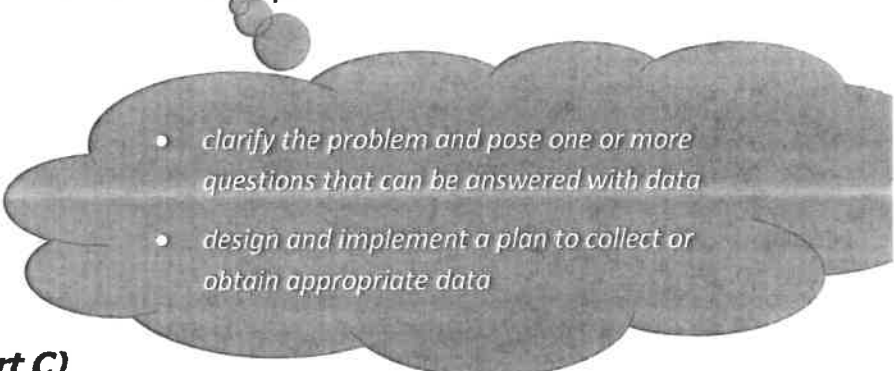
Scenario: A local ice cream shop has been keeping track of how much ice cream they sell versus the temperature at noon that day. Here are their records for the last 2 weeks:

Ice Cream Sales Vs Temperature	
Temperature (°C)	Ice Cream Sales (\$)
14.5	235
16.8	325
12.2	190
10.0	400
18.5	410
22.2	535
19.6	440
25.7	615
23.8	560
18.0	420
17.5	398
24.8	586
28.1	733
19.2	416

Question: Does an increase in daily temperature affect ice cream sales?

Step One: Begin your paragraphs/report

Use the statistical thinking process planning page to start to write your introduction. This will incorporate the **(Part A and B)** of the statistical thinking process

- 
- clarify the problem and pose one or more questions that can be answered with data
 - design and implement a plan to collect or obtain appropriate data

Step Two: Graphical Analysis (Part C)

- Draw a scatterplot of the data. Clearly label all axes (identifying the independent and dependent variables), give an appropriate title and accurately record all data.
- Fit a trend line by eye.

Outliers are any values which appear to be very different from the majority of the data. They are extreme values and may have been caused by an error. They should be removed from your data set and not included in your analysis.

- If outliers are present, remove them. Explain why you have chosen to exclude any data and the effect it would have on your results if it were included.
- **Extrapolation and interpolation:** Use your trend line to make predictions on further data, if it were available, that would support any trends you have observed. Describe some of the potential dangers of extrapolation.

Step Three: Interpretation of the Graph/Data (Part D)

You need to now keep writing your report. The following information will help you with what needs to be included in this section:

When investigating two numerical variables, you will look at how the data is distributed, whether there is a relationship between the two variables, and whether one variable causes the other.

Causation and Association:

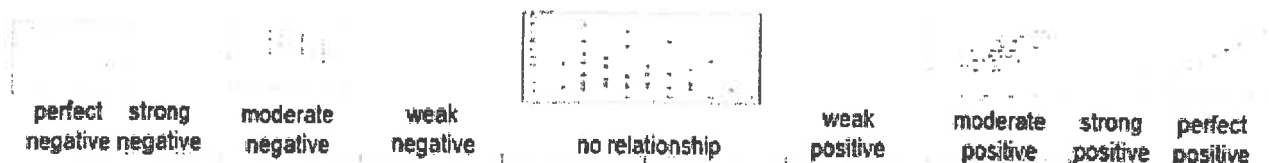
What is causation?

A relationship between two variables, the independent and dependent response, are said to be causal if a change in the independent variable (usually graphed on x axis) actually causes a change in the dependent variable (usually graphed on the y axis).

What is association or correlation?

Both of these words are used to describe relationships that exist between two or more variables. Correlation tends to be used when describing the strength of a relationship between two numerical variables (for example height and arm span).

<i>Strong Negative Correlation</i>	<i>Weak Negative Correlation</i>	<i>No Correlation</i>	<i>Weak Positive Correlation</i>	<i>Strong Positive Correlation</i>
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Add the following into your report (this will form the body of your report):

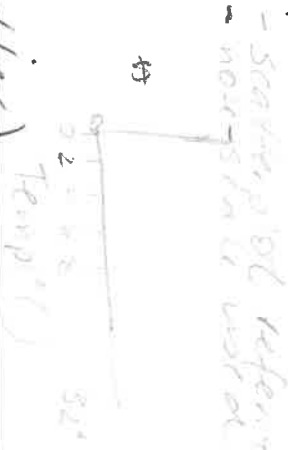
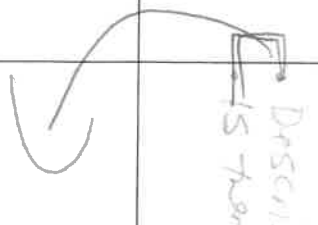
- Examine the linear relationship of your graph (using your trend line). Describe the association between the two variables in terms of direction (positive/negative), form (linear/non linear) and strength (strong/moderate/weak).
- Describe any patterns and features of your graph/data that may suggest the presence of an association.
- Is there a causal relationship between the two variables you investigated? Justify your answer.
- If there is an association, but not a causal relationship between the two variables you investigated, identify any non-causal explanations for the association, such as coincidence.

Step Four: Conclusion (Part E)

- Using all of the information you have gathered, write a conclusion of your findings, using the context of the provided data. Ensure that you have clearly answered the question.

Question: Does an increase in daily temperature affect ice cream sales?

Statistical Thinking Process Planning Sheet

Clarify the problem and pose question/s that can be answered with data	- local shop needs to assess whether or not temperature affects ice cream sales.	
Design and implement a plan to collect or obtain appropriate data	Design - Temp taken @ noon each day. - Data collected in table. - Dependent variable (y-axis) ice cream sales. - Independent variable (x-axis) = temperature.	Implementation - Scatter plot referred to. - non-graphical version 
Analyse information/data by using graphical or numerical techniques	- Draw a scatter plot (why is it an outlier) - Discuss outliers (why is it an outlier) - Show & make a comment on interpolation - Show & make a comment on extrapolation - Describe the association - Is there a causal relationship? - Does 1 variable cause something to the other? - first variable (cause) - second variable (effect)	
Interpret the results and relate the interpretation to the original question		
Communicate findings in a systematic and concise manner.		

Answer/Findings/ Solution:

Year 12 Essential Mathematics Unit 3

Using Bivariate Data

(In-class preparation for Statistical Investigation)

(Syllabus: 3.4.12, 3.4.13, 3.4.15-3.4.19)

TASK:

The teacher will present you with bivariate data to interpret and analyse. You will need to draw scatterplots (with and without technology) and use these to describe any patterns/features with the data. You should then be able to conclude if there is an association or correlation with the data.

Based on what you have learned in class, you will complete a statistical investigation on data provided to you. This will be an assessment done in class under test conditions.

* highlight here - no notes, no assistance + 1 lesson only.

Part One

Step One: Introduction

A local ice cream shop has been keeping track of how much ice cream they sell versus the temperature at noon that day. Here are their records for the last 2 weeks:

don't be too sure

x = independent *y = dependent*

Temperature (°C)	Ice Cream Sales (\$)
14.5	235
16.8	325
12.2	190
10.0	400
18.5	410
22.2	535
19.6	440
25.7	615
23.8	560
18.0	420
17.5	398
24.8	586
28.1	733
19.2	416

Outlier

Question: Does an increase in daily temperature affect ice cream sales?

Step Two: Graphical Analysis

- Draw a scatterplot of the data. Clearly label all axes (identifying the independent and dependent variables), give an appropriate title and accurately record all data. ** Find highest + lowest to get scale*
- Fit a trend line by eye.

Outliers are any values which appear to be very different from the majority of the data. They are extreme values and may have been caused by an error. They should be removed from your data set and not included in your analysis.

- If outliers are present, remove them. Explain why you have chosen to exclude any data and the effect it would have on your results if it were included.

Extrapolation and interpolation: Use your trend line to make predictions on further data, if it were available, that would support any trends you have observed. Describe some of the potential dangers of extrapolation.

estimation of a value based on extending the trend line *estimation between 2 values*

How many ice creams sold at 20°C (interpolation)
** Students do another one*
** How many ice-creams sold at 32°C (extrapolation)* *Do one more*

Step Three: Interpretation of the Graph/Data

When investigating two numerical variables, you will look at how the data is distributed, whether there is a relationship between the two variables, and whether one variable causes the other.

Causation and Association:

What is causation?

A relationship between two variables, the independent and dependent response, are said to be causal if a change in the independent variable (usually graphed on x axis) actually causes a change in the dependent variable (usually graphed on the y axis).

What is association or correlation?

Both of these words are used to describe relationships that exist between two or more variables. Correlation tends to be used when describing the strength of a relationship between two numerical variables (for example height and arm span).

Strong Negative Correlation	Weak Negative Correlation	No Correlation	Weak Positive Correlation	Strong Positive Correlation
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perfect negative

strong negative

moderate negative

weak negative

no relationship

weak positive

moderate positive

strong positive

perfect positive

- Examine the linear relationship of your graph (using your trend line). Describe the association between the two variables in terms of direction (positive/negative), form (linear/non linear) and strength (strong/moderate/weak).
- Describe any patterns and features of your graph/data that may suggest the presence of an association.
- Is there a causal relationship between the two variables you investigated? Justify your answer.
- If there is an association, but not a causal relationship between the two variables you investigated, identify any non-causal explanations for the association, such as coincidence.

Step Four: Conclusion

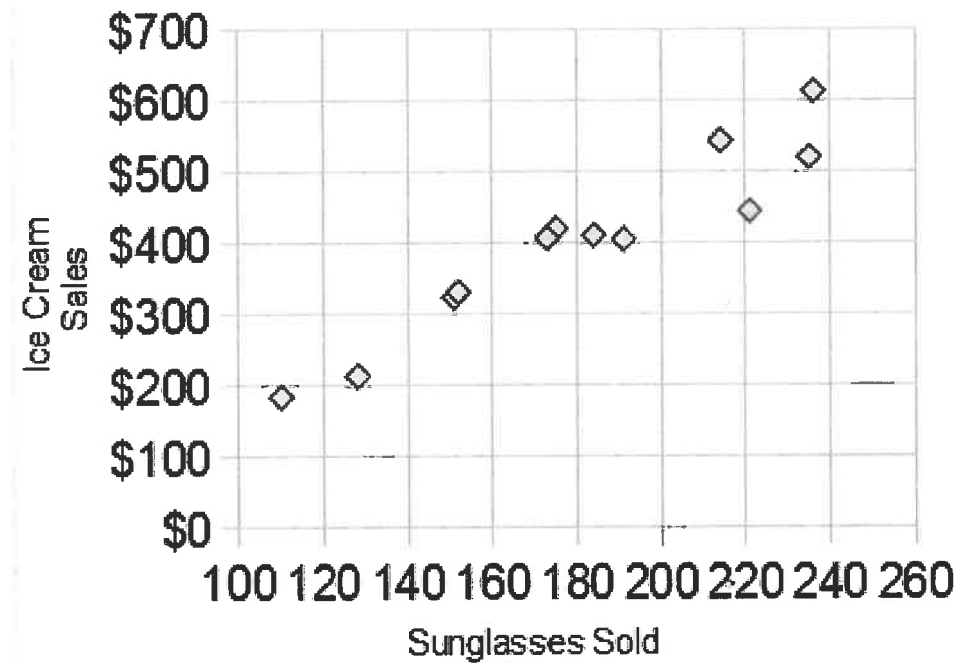
- Using all of the information you have gathered, write a conclusion of your findings, using the context of the provided data. Ensure that you have clearly answered the question.

Question: Does an increase in daily temperature affect ice cream sales?

Part Two:

Introduction

The same ice cream shop found out how many sunglasses were sold by a retail store for each day and compares them to their ice cream sales. This graph shows the results:



Lesson 2.

→ Using the Statistical Investigative Process (you used in Part One), answer this question:

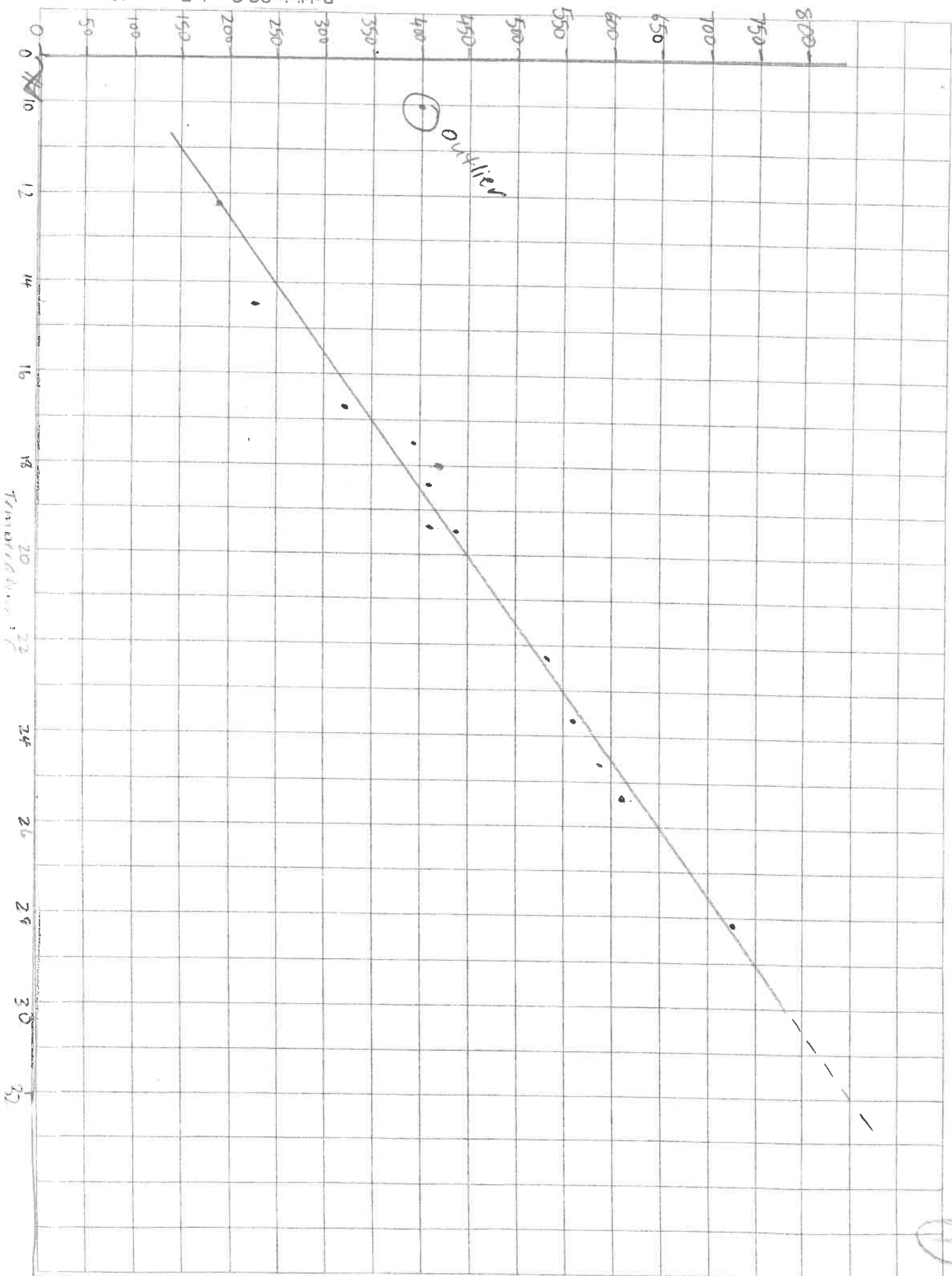
"Does an increase in the number of sunglasses sold affect ice cream sales?"

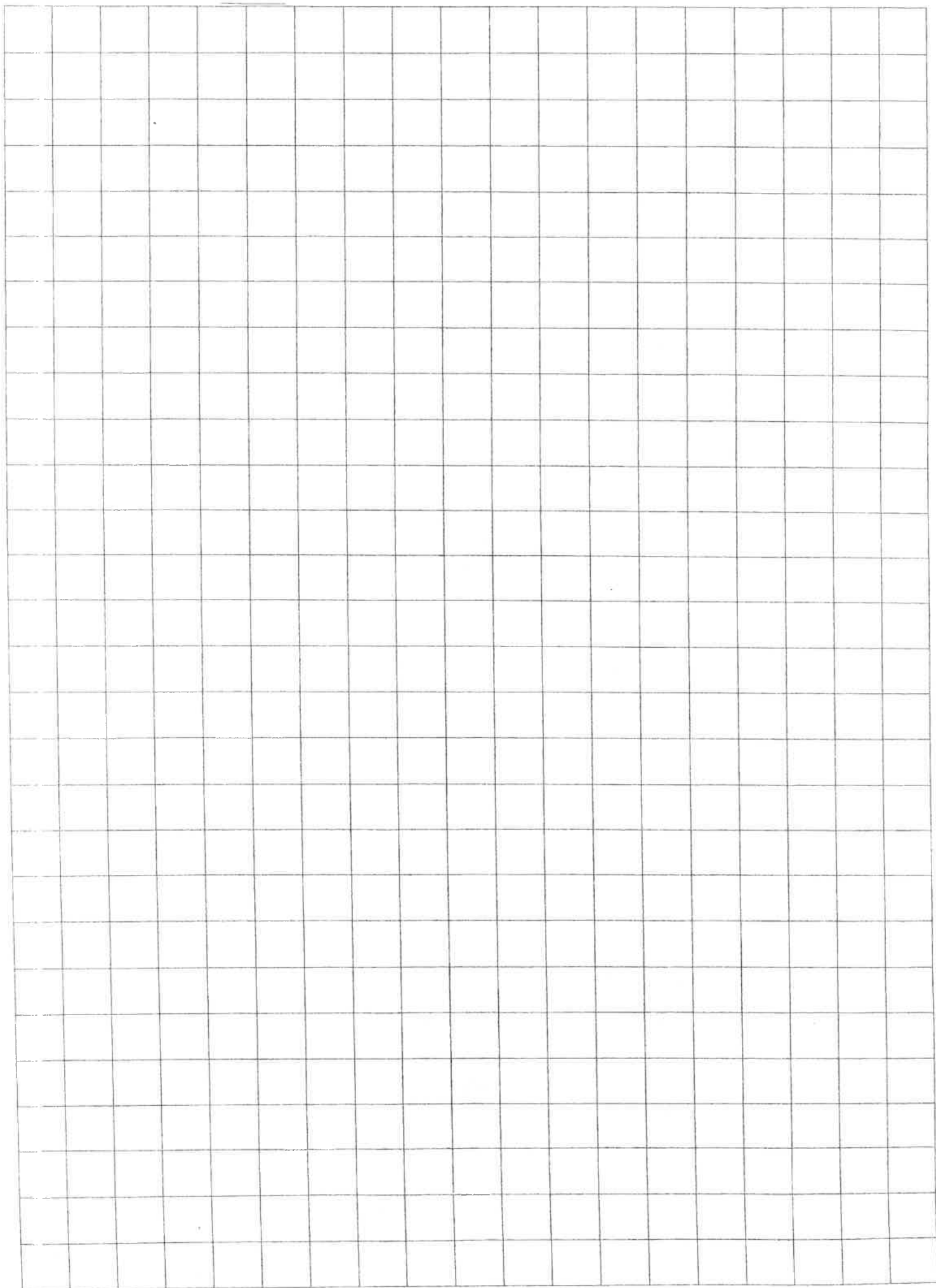
Your analysis should include:

1. Introduction
2. Graphical Analysis
3. Interpretation of the Graph/Data
4. Conclusion

Ice-cream sales (\$)

Baldivis SC Graph Paper - 10mm





The dependant variable is the sales of ice-cream. This value will rely/depend on the day's temperature. This will go on the y-axis

The independant variable is the temperature, as variation does not depend on that of another. It will go on the x-axis

Outliers - There is one outlier in this data set. It is 400 ice creams when it is 10°C . These sales are much bigger than at other similar temperatures. If it was included the value would affect the mean or "average" sales, it would make the average larger.

Interpolation = icecreams @ 20°C approx 450

Extrapolation - at 32°C there are approximately 850 ice-creams.

Extrapolation is an estimation. But unlike interpolation (which is an estimation between 2 points), extrapolation only relies on "known" data on one side. The further you extrapolate the less accurate it is as you need more data to know what the slope is

Causation / Association - The association between temperature + ice-cream sales is positive, linear + strong.

This is because the graph steps up from left-right (positive)

The scatter of the dots is a linear pattern (linear)

These dots form tightly around the line of best fit (strong)

Yes there is a causal relationship between these 2 variables. As the temperature increases so does the sales of ice-creams. Because this line is approximately 45° what this means that both variables increase by the same amount.

Conclusion: In conclusion the graph is showing the relationship between temperature + ice-cream sales. It is directly related because as the temperature increases so does the sales of ice-cream. The graph has formed a positive, linear + strong correlation. This graph shows that yes, an increase in temperature does affect ice-cream sales.